

TITLE PAGE

Problem Statement ID <<FILL IN>>

Problem Statement Title Eco-Loop Building Agents — autonomous closed-loop HVAC control using EnergyPlus, MCP and a local open-source LLM

Theme <<FILL IN>>

PS Category Software

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EnergyPlus 26.1 · pyenergyplus · FastMCP · Model Context Protocol · Ollama + Llama 3.2 3B · Pydantic · Streamlit

A building that runs itself.

EnergyPlus supplies the physics, a local open-source LLM supplies the judgement, and the Model Context Protocol carries every reading and every command between them — while the simulation is still running.

-8.5%

HVAC energy, summer week

336 / 336

agent turns — zero failures

100%

open source, runs on a laptop

IDEA TITLE

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Proposed Solution

Traditional BMS follow rigid clock schedules. Eco-Loop turns the building into a self-correcting agent.

Detailed explanation

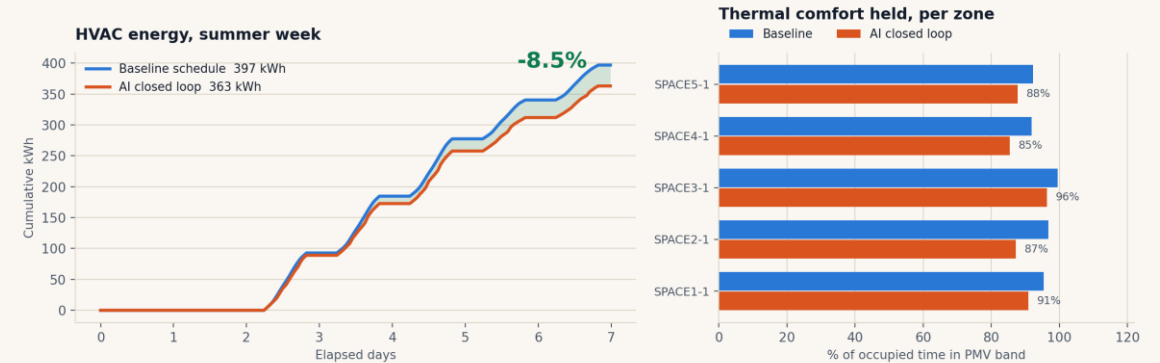
- EnergyPlus runs inside the Python process; sensors read every zone timestep
- Every 60 simulated minutes the aggregated state goes to a local LLM through six MCP tools
- The model reasons over comfort, demand and grid carbon, then returns setpoints

How it addresses the problem

- Control adapts continuously to weather, occupancy and carbon intensity instead of a fixed schedule

Innovation and uniqueness

- Setpoints are injected into the LIVE solver via the actuator API — no IDF rewrite, no restart, so the loop closes inside one simulation
- MCP is the safety boundary: every command is validated server-side before it can reach an actuator



Measured, not projected

8.5% less HVAC energy across a summer week, while holding occupants closer to thermal neutrality than the baseline (mean PMV $-0.43 \rightarrow -0.02$).

Why it is different

Most LLM-and-simulation work rewrites the model file and re-runs it — that can only compare finished runs. It cannot react to a zone drifting out of comfort at 14:00 on day 3.

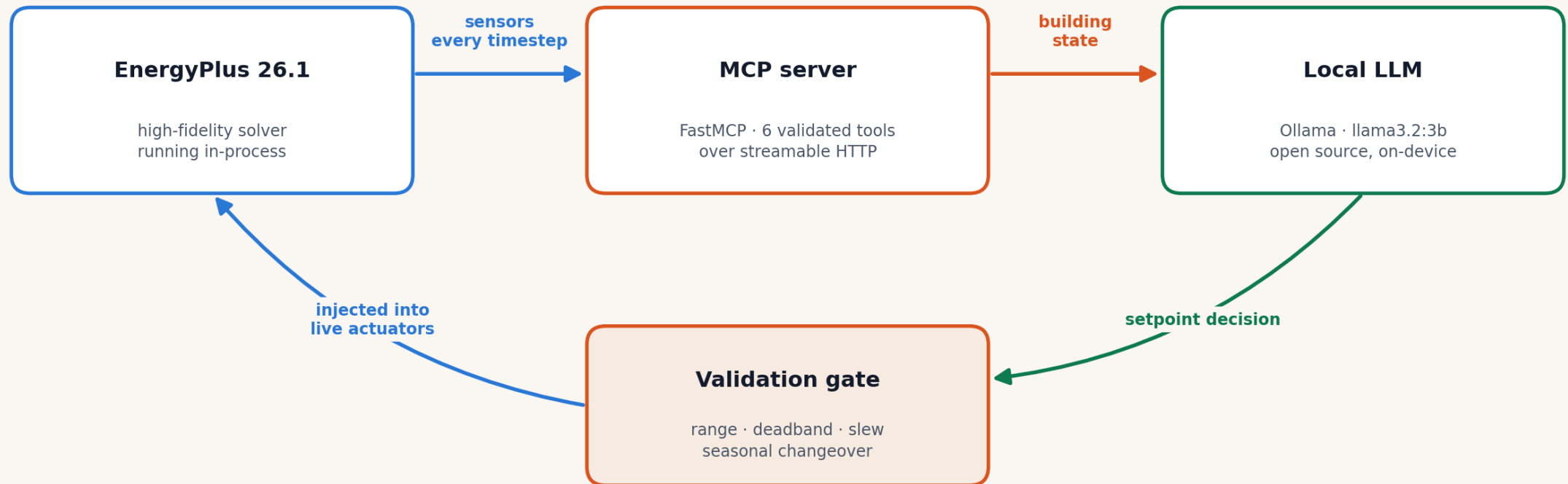
TECHNICAL APPROACH

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Technologies EnergyPlus 26.1 (pyenergyplus C API) · Python · FastMCP over streamable HTTP · Ollama running Llama 3.2 3B · Pydantic validation · Streamlit + Plotly dashboard

Methodology sample every timestep → aggregate per control interval → LLM tool-calling turn → validate → inject into live actuators

CLOSED LOOP · no file rewrite, no restart · 168 agent decisions per simulated week



FEASIBILITY AND VIABILITY

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336 / 336

agent turns — 0 fallbacks, 0 timeouts, 0 actuation errors

−8.5%

HVAC energy, summer week (−3.7% winter)

2.1 s

median agent decision latency, on-device

Feasibility

- Runs end to end on one consumer laptop — 4 GB VRAM, no cloud, no API cost
- Fully open source: EnergyPlus, Llama 3.2, FastMCP

Potential challenges and risks

- LLM latency inside a blocking simulation callback
- A small model can move the wrong setpoint, or run heating and cooling against each other
- One setpoint pair cannot satisfy five differently-loaded zones

Strategies for overcoming them

- Interval batching + a hard per-turn deadline; on timeout the loop holds the last accepted command rather than stalling
- Observations state the required direction, so the model cannot invert the physics
- Seasonal changeover parks the idle mode's setpoint — heating and cooling can never fight
- A per-zone proportional trim corrects each zone from its own PMV

Viability at scale

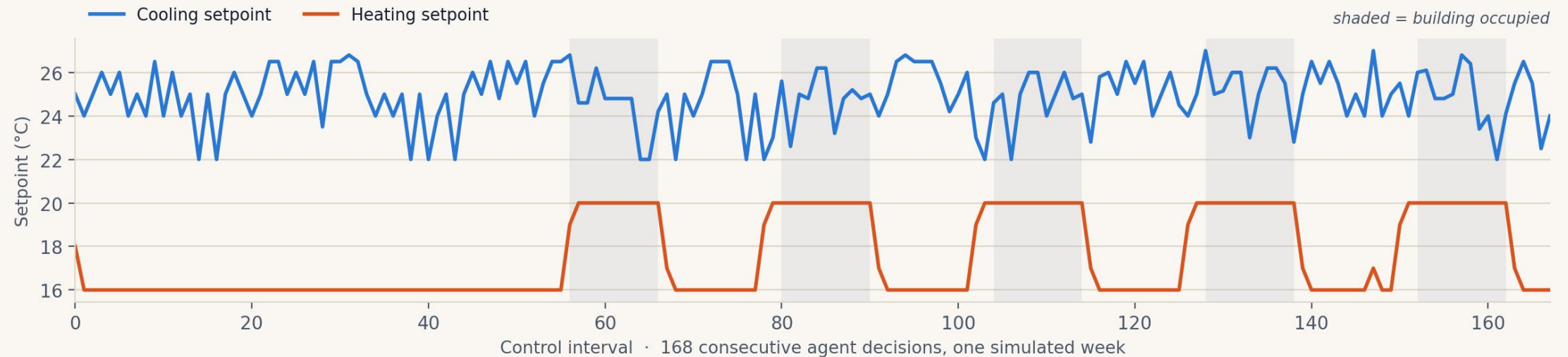
Buildings consume roughly 40% of global energy, so a single-digit HVAC reduction is material at portfolio scale. The loop is model-agnostic — any EnergyPlus model with a thermostat works — and swapping the cognitive engine for a larger local or hosted model needs no change to the MCP tool layer.

ARTIFACTS

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Code github.com/ayushYadav1107/Eco_loop

Baseline and runtime-generated .idf models, the full agent decision trail (agent_decisions.jsonl), per-timestep EnergyPlus results and the savings dashboard are all committed — a fresh clone reproduces these figures.



The agent's own stated reasoning, taken verbatim from agent_decisions.jsonl

07-17 09:00 *"Raise cooling_sp to about 27.0 C this turn — this fixes comfort AND saves energy"*

07-17 11:00 *"HOLD the current setpoints. Do not trim energy here — you would push occupants out of the band"*

07-17 12:00 *"Nudge cooling_sp up to save energy while keeping PMV within the band"*

RESEARCH AND REFERENCES

Simulation and building physics

- EnergyPlus Engineering Reference and EMS Application Guide — energyplus.net
- pyenergyplus Data Exchange API — `get_variable_handle`, `set_actuator_value`
- ISO 7730:2005 and ASHRAE Standard 55 — PMV/PPD thermal comfort model

Agent, protocol and inference

- Model Context Protocol specification — modelcontextprotocol.io
- FastMCP — gofastmcp.com
- Meta Llama 3.2 served locally through Ollama — ollama.com

Project

- Repository, architecture document and reproducible results — github.com/ayushYadav1107/Eco_loop

Reproduce the result

`python main.py prepare`

`python main.py run-baseline --start 07-15 --end 07-21`

`python main.py run-ai --start 07-15 --end 07-21`

`python main.py dashboard`

Temperature 0, so repeated runs return identical totals — the comparison is auditable rather than anecdotal.

Honest limitation

One setpoint pair drives five differently-loaded zones. Per zone the agent holds 85–96% of occupied time inside the PMV band; the stricter worst-zone metric is lower.

Per-zone setpoints are the next step — the actuator handles and per-zone PMV are already in place.